# spot chekers lis alt proof

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**Lemma 0.1.** *an element $i$ being 0 big can be detected with probability $\frac{3}{4}$ in $O(\log n)$ time.*

*Proof.* wlog let $j$ be the smallest witness to $i$'s bigness and $j > i$. There exists some $n$ such that $j \in [i + 2^n, i + 2^{n+1}]$. Since $j$ is the smallest witness, the number of violations that $i + 2^n$ sees is $< \frac{1}{2}$. the number of violations that $i + 2^{n+1}$ witnesses is at least $\frac{j-i+1}{2} \geq \frac{1}{4} \cdot 2^{n+1}$. The probability of finding a violation given $n$ samples is $1 - \left(\frac{3}{4}\right)^n$. If $n$ is large enough, the probability that arbitrarily picking an element $k$ in $[i + 2^n, i + 2^{n+1}]$ a violation $f(k) < f(i)$ is found is greater than $\frac{3}{4}$. Doing this procedure on $i$ for all $n$ in both directions until indecies exceed the bounds of the array guarantees that this algorithm returns BIG in $2 \log_2 n$ time with at least $\frac{3}{4}$ probability. □

**Lemma 0.2.** *Given an array of booleans of length n with $\varepsilon$ Trues, the probability of picking at least 1 True given $m$ samples of the uniform distribution is $\geq 1 - (e^{-m\varepsilon/2})$.* [1]

*Proof.* We know

$$P(X \leq (1-\delta)\mu) \leq e^{-\frac{\mu\delta^2}{2}} \tag{1}$$

by the Chernoff bound formula. The expected number of times True is picked given $m$ samples is $m\varepsilon$. By setting $\delta = 1$ and $X$ to be the sum of $m$ outcomes of Bernoulli distribution with success $\varepsilon$, we see the probability of none of the Bernoulli distributions returning 1 (i.e. sampling true) is

$$P(X = (1-1)\mu = 0) \leq e^{-\frac{m\varepsilon 1^2}{2}} \tag{2}$$

The probability of picking at least 1 True is 1 minus this value.[2]

□

**Lemma 0.3.** *Given $a < b \in [n]$ and $f(a) > f(b)$, either $a$ or $b$ is 0 big.*

*Proof.* let $c \in (a, b)$. $f(c)$ is either

1. greater than $f(a)$

[1] this is independent of $n$ (!!)

[2] The RHS of the can be set arbitrarily small by setting $m = O\left(\frac{1}{\varepsilon}\right)$.

2. less than $f(a)$ and greater than $f(b)$

3. less than $f(b)$

We count the number of violations $a$ witnesses for $b$, and $b$ witnesses for $a$. The proof would follow if either $a$ is the bigness witness for $b$ or $b$ is the bigness witness for $a$. Respective to the enumeration, $c$ is in violation of $b$, $a$ and $b$, $a$ in each case respectively. Since every $c$ is assigned as a violation for either $a$ or $b$, by pigeonhole there must be at least $\frac{b-a+1}{2}$ violations for either $a$ or $b$. □

# 1 The algorithm

Given $f : [n] \to \mathbb{R}$

**for** l **do** $O(\frac{1}{\varepsilon})$ times:

  pick element $a \in [n]$ and check if $a$ is 0 big using lemma 0.1

  **if** 0 big **then**

    return FAIL

return PASS

The probability detecting an element $a$ is 0 big given it is in fact 0 big should be set to be $\frac{3}{4}i$ and the number of times the for loop repeats should be set so that the probability of sampling a 0 big element is at least $i$.

If $\varepsilon n$ elements must be removed for the function $f$ to be monotone, then there must exist at least $\varepsilon n$ 0 big elements. This is true because an element $e$ in violation of monotonicity implies $e$ either works as $a$ or $b$ in lemma 0.3.

$$\begin{aligned}
&= P(\text{FAIL}|\varepsilon \text{ violations of monotonicity}) \\
&= P(\text{FAIL}|\varepsilon \text{ 0 big elements}) \\
&= P(\text{FAIL}|\text{test 0 big element}) \cdot P(\text{test 0 big element}|\varepsilon \text{ 0 big elements})
\end{aligned} \tag{3}$$

The final line is equal to greater than $\frac{3}{4}$.

NOTES: ignore

$m\varepsilon$ is the expected value

0 big can be checked in $\log n$ time. candidate $2^n$ being witness for i being 1/4 big is checked in O(1) (this overfails but thats ok) assume that $i$ is 0-big. check s elements in interval $[i, i+2^n]$. the probability that a witness $j$ in the interval $[i+2^{n-1}, i+2^n]$ is NOT found is $(3/4)^s$. making s large makes the probability that the algo fails to find 0-big very small. note $s$ is $O(1)$.

given the sequence should fail, the expected number of violations sampled is 1.

P(output F| bad sequence) = 3/4

P(output F | test bad element) $= 1 - \delta$ P(test bad element | bad sequence) $= \varepsilon$

P(A) = P(A|B) P(B)

assume: $\varepsilon$ violate monotonicity

check $1/\varepsilon$ times. check for 0 big. if not found, continue. else fail.

pass

**Lemma 1.1.** *if an element $i$ is in a subset $s$ such that when $n - s$ gives an optimal monotone subsequence, then $i$ is 0 big.*

*Proof.* $i$ must have at least one witness $j$ such that $i < j, f(i) > f(j)$ or $i > j, f(i) < f(j)$. □

**Lemma 1.2.** *given an element in a list length $n$ and that is $\varepsilon_f n$ away from monotonicity there exists an algorithm detects its violation from monotonicity that runs in $\log n$ time*

*Proof.* □